

Pre-training for Speech Translation: CTC Meets Optimal Transport

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Background: Speech-to-Text Translation Is Transforming Communication

- We enable real-time cross-lingual communication for captioning, conferencing, and accessibility.
- We pursue end-to-end ST to lower latency and reduce errors versus cascaded ASR→MT.
- We scale multilingual models to deploy quickly across many language pairs.

Our Problem: We Bridge the Speech–Text Modality Gap in End-to-End ST

- We face length, structure, and timing mismatches between speech and text, which complicate alignment.
- We observe that standard pre-training often discards alignment signals.
- Our goal is to reduce the modality gap during pre-training without changing the ST architecture.

Why CE-Based Pre-training Falls Short for Us

- For ASR (CE): we learn alignment mostly in the decoder; ST discards it, so alignment is lost.
- For MT (CE): the encoder is not reused by speech, so learned alignments don't reach the speech encoder.
- We therefore need alignment directly in speech representations.

Core Idea: We Make Alignment a Pre-training Objective

- We use CTC to teach the speech encoder monotonic alignments to text.
- We add Siamese pre-training to align speech and text encoders via optimal transport (OT).
- We do not change the downstream ST architecture; our recipe is drop-in.

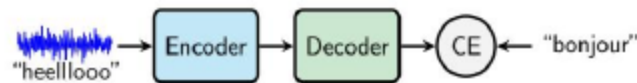


Figure: Illustration of our pre-training setup.

We Use CTC-Only ASR to Embed Alignment in the Encoder

- We leverage CTC to align frames to tokens via monotonic paths without a decoder.
- We keep the alignment signal inside the encoder to transfer it to ST.
- We find this is empirically faster and more effective for ST than CE-only.

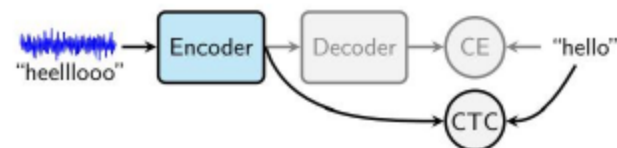


Figure: Illustration of CTC-only ASR pre-training (no CE branch).

We Align Speech and Text Encoders with OT (Siamese Pre-training)

- We use two encoders (speech/text) that produce sequences in a shared dimension.
- We apply Wasserstein OT to align variable-length sequences via minimal transport.
- Our objective is $\text{CTC}(\text{speech}) + \alpha \cdot \text{OT}(\text{speech}, \text{text})$, with $\alpha \approx 0.1$.
- We choose OT over simpler distances based on consistent dev-set gains.
- We optionally initialize the text encoder from MT to provide a target space; we see gains with or without this.

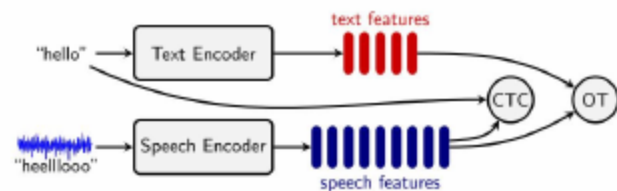


Figure: Illustration of Siamese pre-training with optimal transport.

OT Details: Efficient Sinkhorn and a Monotonic Bias

- We use entropically-regularized OT with GPU-friendly Sinkhorn iterations.
- We append normalized positions to features and include them in the OT cost.
- We enforce soft monotonicity efficiently by appending positions.

- We evaluate on MuST-C (8 En→X European targets) and CoVoST-2 (En→X, X→En).
- We use BLEU for ST and WER to analyze ASR pre-training.
- We focus on multilingual settings and avoid external data unless noted in baselines.

- We use encoder-decoder Transformers (medium d=512, large d=1024).
- We compare CE pre-training, CTC, CTC+CE, and our Siamese-PT (CTC+OT).
- We apply Siamese-PT on top of vanilla and multi-task training frameworks.

We Show CTC > CE; CTC+OT Yields the Largest Gains

- We find CTC beats CE consistently; Siamese-PT (CTC+OT) yields the largest gains.
- We observe improvements with and without MT pre-training; multilingual averages improve.
- We see broad relative BLEU gains across MuST-C and CoVoST-2.

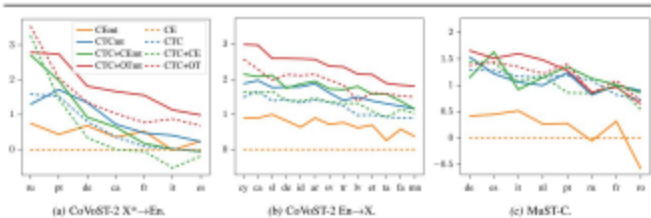


Figure: Key development results compared across pre-training variants.

We Validate: CTC > CE; CTC+OT Achieves the Best BLEU

- We improve BLEU with CTC over CE despite higher WER, indicating better encoder alignment.
- We achieve the best BLEU with Siamese-PT (CTC+OT) at modest training-time overhead.
- We conclude that alignment quality, not ASR WER, predicts ST success.

Table: BLEU and WER (in parentheses) on MuST-C dev sets.

Method	En→De	En→Fr	Training time*
CE	23.04 (13.1)	28.89 (12.6)	7.10h
CTC	24.08 (17.7)	29.91 (17.3)	6.27h
CTC+CE	24.28 (12.9)	30.21 (11.9)	7.78h
CTC+OT	24.74 (17.6)	30.31 (17.1)	7.62h

- We achieve 29.8 avg BLEU on MuST-C (large, multilingual), +1.0 over CE and +0.6 over CTC baselines.
- We see consistent gains across most target languages.
- Under no-external-data, we set a new state of the art and remain competitive with recent systems that use extra data.

Generalization and Practicality

- We raise averages on CoVoST-2 for both En→X and X→En over strong baselines.
- We stack on recent multi-task systems and obtain further gains.
- We train efficiently: CTC is faster than CE; CTC+OT has modest overhead (~1.22× vs CTC).

- We note CTC still has higher WER than CE; OT adds moderate compute cost.
- We are sensitive to transcript quality and tokenization choices.
- We commit to privacy, consent, and secure data handling in deployments.

Why It Works: We Find That Alignment—Not ASR WER—Drives ST

- We preserve monotonic token alignments in the CTC encoder that transfer to ST.
- We reduce cross-modal discrepancy with OT beyond ASR accuracy improvements.
- We thereby create translation-friendly speech features before ST.

Conclusion: We Deliver Simple, Architecture-Agnostic Gains for ST

- We reframe the modality gap as a pre-training problem solved via CTC and OT.
- We achieve state-of-the-art results without external data and boost multi-task systems.
- We offer a drop-in recipe for existing ST toolkits with consistent, reliable improvements.

Questions & Discussion

- Thank you for your attention!
- We welcome your questions and feedback.

Future Work and Impact

- We plan to leverage external ASR/MT resources for very low-resource languages.
- We will explore richer transport costs, positional encodings, and broader tasks.
- We aim to extend our approach to speech retrieval and spoken language understanding.

Acknowledgments

- We thank our collaborators, supporting institutions, and the open-source community.
- We acknowledge responsible AI practices and community benchmarks enabling fair comparison.